

Venkatesh Lakkakula

📞 (+91) 965-257-8726

✉ lakkakula.venkatesh8726@gmail.com

🌐 LinkedIn

🐙

GitHub

Professional Summary

Software Engineering Senior Analyst with 5 years of experience architecting scalable web applications and distributed systems. Polyglot engineer proficient in Ruby on Rails, the MERN Stack, and Java. Strong advocate for Test Driven Development (TDD) and rigorous testing practices to minimize technical debt. Proven track record in HealthTech and SaaS, driving revenue growth by 32% through algorithmic optimization and feature delivery.

Technical Skills

Languages: Ruby, JavaScript (ES6+), TypeScript, Java, Python, SQL

Backend: Ruby on Rails, Node.js, ExpressJS, Spring Boot, Sidekiq, FastAPI, Uvicorn

Frontend: ReactJS, Redux Toolkit, NextJS, TypeScript, Tailwind CSS, Bootstrap, HTML5, CSS3

Databases & Cloud: PostgreSQL (PostGIS), MongoDB, Redis, AWS (EC2, S3), Docker, Git

Core Competencies: System Design, Real-Time Systems (WebSockets), API Design, Test Driven Development (TDD), CI/CD, AI Workflow Orchestration

Professional Experience

Evernorth Health Services (Cigna Group)

Software Engineering Senior Analyst

Feb 2025 – Present

Hyderabad, India

- Architected and developed the **Mirth Eligibility Dashboard**, providing administrators with visibility into file uploads and patient data flow.
- Led migration of legacy Crystal Reports to a modern web platform, streamlining billing workflows for the Finance team.
- Championed Test Driven Development, achieving **100% code coverage** across more than **70% of core application models**.
- Enhanced EDI 271 response processing, improving data accuracy for downstream research teams.
- Successfully deprecated legacy functionality and executed complex Rake tasks to remove obsolete records without regressions.

Geosnapshot

Full Stack Engineer / React Engineer

Dec 2020 – Jan 2025

Remote

- Engineered a Smart Tagging algorithm that significantly improved image search precision and retrieval speed.
- Designed and delivered a core SaaS feature that directly increased platform revenue by **32%**.
- Reduced dashboard load times by **40%** through server-side pagination and frontend rendering optimizations.
- Modernized the application's UI/UX, improving accessibility and overall user experience.
- Established robust TDD practices and conducted code reviews to maintain engineering quality.
- Developed a Showreel generation feature using user selfies, increasing average session duration by **41%**.

Key Technical Projects

AI Engineering & Workflow Orchestration

FastAPI, Python, Vanilla JavaScript, HTML5, CSS3

- Engineered a FastAPI monitoring dashboard for an autonomous LLM job-sourcing agent automating job scraping, candidate evaluation, resume generation, and Slack notifications.
- Designed a runtime telemetry layer by intercepting Python's print stream, enabling real-time workflow monitoring without modifying pipeline logic.
- Leveraged FastAPI BackgroundTasks for asynchronous execution of long-running scraping and LLM evaluation processes.

- Built a lightweight JSON-based persistence layer with robust error handling for execution history and metrics.
- Developed a responsive glassmorphic dashboard with custom SVG assets and real-time polling endpoints for live console updates.

Real-Time Collaborative Dashboard

MERN Stack, Socket.io, Redux Toolkit, MongoDB

- Built a collaborative Kanban board using Socket.io, synchronizing task movements instantly across connected clients.
- Implemented Redux Toolkit for predictable state management across complex SPA components.
- Designed optimized MongoDB schemas and implemented stateless JWT authentication to simulate a microservices architecture.

Education

RVR & JC College of Engineering

Bachelor of Technology in Electrical & Electronics Engineering

Jun 2016 – Dec 2020

Guntur, Andhra Pradesh

Certifications

- Full Stack Web Development Bootcamp – Newton School